



12761 - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionization at $z > 8$

Cycle: 5, Proposal Category: DD

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Julien Anton Wolf (PI) (ESA Member)	Max Planck Institute for Astronomy
Francesco Guarneri (CoI) (ESA Member) (CoPI)	Universitat Hamburg
Daming Yang (CoI) (ESA Member) (CoPI)	Universiteit Leiden
Eduardo Banados (CoI) (ESA Member)	Max Planck Institute for Astronomy
Prof. Joseph Hennawi (CoI) (US Admin CoI)	University of California - Santa Barbara
Dr. Silvia Belladitta (CoI) (ESA Member)	Max Planck Institute for Astronomy
Prof. Anna-Christina Eilers (CoI)	Massachusetts Institute of Technology
Dr. Jan-Torge Schindler (CoI) (ESA Member)	Universitat Hamburg
Dr. Raphael Erik Hviding (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Roberto Decarli (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Prof. Masafusa Onoue (CoI)	Waseda University
Dr. Frederick Davies (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Daniel J Mortlock (CoI) (ESA Member)	Imperial College London

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Spectroscopy				
	1	High-resolution spectroscopy G140H/G235H	NIRSpec Fixed Slit Spectroscopy	(1) J1504+6252
	2	Medium-resolution spectroscopy G395M	NIRSpec Fixed Slit Spectroscopy	(1) J1504+6252
Imaging				

Folder	Observation	Label	Observing Template	Science Target
	3	NIRCAM pointing F150W/F356W	NIRCam Imaging	(1) J1504+6252
	4	NIRCAM pointing F200W/F444W	NIRCam Imaging	(1) J1504+6252

ABSTRACT

The first quasar known beyond $z = 8$ was discovered after the previous JWST deadline. It was identified in newly acquired Euclid imaging, confirmed by HST/WFC3 slitless spectroscopy, and independently validated with deep LBT/LUCI follow-up. However, telluric absorption and the prohibitive ground-based exposure time required to reach diagnostic signal-to-noise prevent a robust physical interpretation of the rest-UV spectrum. Only JWST can provide the dataset needed to unlock the key physical diagnostics of the central supermassive black hole (SMBH) and reionization.

Unlike the much fainter JWST broad-line objects now found at these cosmic epochs, J1504+6252 unambiguously traces luminous SMBH accretion. It is the first $z > 8$ source for which a reliable single-epoch black-hole mass is within reach. We request NIRSpec fixed-slit spectroscopy with G140H/F100LP, G235H/F170LP, and G395M/F290LP, complemented by NIRCam F150W, F200W, F356W, and F444W imaging. These data will measure the black-hole mass and accretion properties of the most distant known SMBH, enable the first reionization and quasar-duty-cycle constraints from a $z > 8$ quasar sightline, probe early intergalactic metal enrichment, and constrain SMBH–host co-evolution in the first 630 Myr of cosmic history.

The observations are highly time-sensitive: waiting for the next regular JWST cycle would delay the first physically interpretable spectrum of J1504+6252 by at least a year. JWST can open this cosmic frontier now, turning the first known $z > 8$ quasar from a landmark discovery into a definitive benchmark for early SMBH growth and reionization.

OBSERVING DESCRIPTION

This program will obtain the first science-ready JWST characterization dataset for the first known quasar beyond $z=8$. We will carry out NIRSpec fixed-slit spectroscopy with G140H/F100LP, G235H/F170LP, and G395M/F290LP to cover the quasar from the rest-frame UV to the rest-frame optical. The high-resolution rest-UV spectrum will enable reionization and quasar-duty-cycle constraints from the Ly region, measure metal absorption along the first $z>8$ quasar sightline, and characterize the broad UV emission lines. The G395M spectrum will provide the systemic redshift from [O III] and the H-based single-epoch black-hole mass, making this the most distant unambiguous SMBH for which a standard virial mass estimate is within reach.

We will complement the spectroscopy with NIRCам imaging in F150W, F200W, F356W, and F444W. After subtracting the unresolved nuclear emission with a 2D PSF+galaxy decomposition method, these images will search for resolved host-galaxy light from the rest-UV to the Balmer/4000 Å region. The resulting host photometry will provide the first stellar-mass constraint, or stringent upper limit, for a luminous quasar beyond $z=8$, and will test whether the black hole is already overmassive relative to its galaxy at the earliest accessible epoch of quasar growth. Together, the NIRSpec and NIRCам observations will transform the frontier discovery into a benchmark for reionization, early metal enrichment, SMBH growth, and SMBH–host co-evolution.

Proposal 12761 - Targets - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionization at z...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	J1504+6252	RA: 15 04 52.2914 (226.2178808d) Dec: +62 52 24.39 (62.87344d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000	
Comments: Category=Galaxy Description=[Active galactic nuclei, High-redshift galaxies, Quasars] Extended=NO					

Proposal 12761 - Observation 1 - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionizati...

Observation	Proposal 12761, Observation 1: High-resolution spectroscopy G140H/G235H										Fri Jun 12 14:00:37 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	J1504+6252	RA: 15 04 52.2914 (226.2178808d)			Proper Motion RA: 0						
			Dec: +62 52 24.39 (62.87344d)			Proper Motion Dec: 0						
			Equinox: J2000			Epoch of Position: 2000						
		Comments: Category=Galaxy Description=[Active galactic nuclei, High-redshift galaxies, Quasars] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	FULL	F140X	NRSRAPID	3	1	1	42.947	304938	
Template	HFF Readout Mode			Slit				Subarray				
	false			S200A1 and S200A2				FULL				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	S200A1	NRSIRS2RAPID	56	1	1	NONE	2	2	1663.133	304938
	2	G140H/F100LP	S200A2	NRSIRS2RAPID	56	1	2	NONE	2	2	1663.133	
	3	G235H/F170LP	S200A2	NRSIRS2RAPID	42	1	3	NONE	2	2	1254.645	
	4	G235H/F170LP	S200A1	NRSIRS2RAPID	42	1	4	NONE	2	2	1254.645	304938

Special Requirements	Group Observations 1, 2, Non-interruptible
----------------------	--

Proposal 12761 - Observation 2 - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionizati...

Observation	Proposal 12761, Observation 2: Medium-resolution spectroscopy G395M										Fri Jun 12 14:00:37 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	J1504+6252	RA: 15 04 52.2914 (226.2178808d)			Proper Motion RA: 0						
			Dec: +62 52 24.39 (62.87344d)			Proper Motion Dec: 0						
			Equinox: J2000			Epoch of Position: 2000						
	Comments: Category=Galaxy Description=[Active galactic nuclei, High-redshift galaxies, Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	FULL	F140X	NRSRAPID	3	1	1	42.947	304938	
Template	HFF Readout Mode			Slit				Subarray				
	false			S200A2				FULL				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	NONE						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395M/F290LP	S200A2	NRSIRS2RAPID	56	1	1	NONE	1	1	831.567	304938

Special Requirements	Group Observations 1, 2, Non-interruptible
----------------------	--

Proposal 12761 - Observation 3 - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionizati...

Observation	Proposal 12761, Observation 3: NIRCAM pointing F150W/F356W									Fri Jun 12 14:00:37 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	J1504+6252	RA: 15 04 52.2914 (226.2178808d)			Proper Motion RA: 0				
			Dec: +62 52 24.39 (62.87344d)			Proper Motion Dec: 0				
			Equinox: J2000			Epoch of Position: 2000				
	Comments: Category=Galaxy Description=[Active galactic nuclei, High-redshift galaxies, Quasars] Extended=NO									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				12
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F356W	SHALLOW4	7	1	12	12	4380.602	
Special Requirements	Fiducial Point Override NRCB1_FULL									

Proposal 12761 - Observation 4 - Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionizati...

Observation	Proposal 12761, Observation 4: NIRCAM pointing F200W/F444W									Fri Jun 12 14:00:37 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	J1504+6252	RA: 15 04 52.2914 (226.2178808d)			Proper Motion RA: 0				
			Dec: +62 52 24.39 (62.87344d)			Proper Motion Dec: 0				
			Equinox: J2000			Epoch of Position: 2000				
	Comments: Category=Galaxy Description=[Active galactic nuclei, High-redshift galaxies, Quasars] Extended=NO									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Detector B1 (compact source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				12
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F200W	F444W	SHALLOW4	7	1	12	12	4380.602	
Special Requirements	Fiducial Point Override NRCB1_FULL									